

PharmaZen

Software Requirements Specification

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Contents

1	Introduction	1
1.1	Purpose	1
1.2	Intended Audience	1
1.3	Intended Use	1
1.4	Product Scope	2
1.5	Risk Definitions	2
2	Overall Description	3
2.1	User Classes and Characteristics	3
2.2	User Needs	3
2.2.1	Registered Customers	3
2.2.2	Pharmacists / Staff	3
2.2.3	System Administrators	3
2.3	Operating Environment	4
2.4	Constraints	4
2.5	Assumptions	5
3	Requirements	6
3.1	Functional Requirements	6
3.1.1	User Registration & Authentication	6
3.1.2	Product Catalog & Search	6
3.1.3	Prescription Management	6
3.1.4	Shopping Cart & Checkout	7
3.1.5	Order Management	7
3.1.6	Admin Dashboard	7
3.2	Non-Functional Requirements	9
3.2.1	Performance	9
3.2.2	Security	9
3.2.3	Reliability & Availability	9
3.2.4	Usability & Accessibility	10
3.2.5	Maintainability & Compliance	10

SECTION 1 · INTRODUCTION

1 Introduction

This document establishes the complete Software Requirements Specification (SRS) for the **PharmaZen** website — an online pharmacy and health-product platform. It defines all functional and non-functional requirements to guide the development, testing, and delivery lifecycle.

1.1 Purpose

The purpose of this SRS is to provide a detailed and agreed-upon description of the PharmaZen website system. It serves as the foundational reference for all project stakeholders — including developers, designers, QA engineers, and business owners — ensuring a shared understanding of what the system must accomplish.

1.2 Intended Audience

This document is intended for the following parties:

- **Development Team** — Front-end and back-end engineers responsible for building the system
- **UI/UX Designers** — Responsible for designing user interfaces that satisfy the stated requirements
- **QA Engineers** — Responsible for validating implemented features against this specification
- **Project Managers** — Overseeing schedule, scope, and delivery milestones
- **Business Stakeholders** — PharmaZen owners and executives reviewing product direction
- **Regulatory Reviewers** — Ensuring compliance with pharmaceutical and data regulations

1.3 Intended Use

This SRS shall be used as:

- A contractual reference between the development team and business stakeholders
- A baseline for test plan development and acceptance criteria
- A guide for identifying and managing change requests throughout the project lifecycle
- A reference document for onboarding new team members

1.4 Product Scope

PharmaZen is a comprehensive e-commerce website specializing in pharmaceutical products, health supplements, personal care items, and related medical goods. The platform will enable customers to browse products, upload prescriptions, place orders, and manage their health profiles.

Core capabilities of the platform include:

- Online pharmacy storefront with product catalog and search
- Prescription upload, verification, and management
- Secure user registration, authentication, and profile management
- Order placement, real-time tracking, and order history
- Admin dashboard for inventory, order, and user management
- Integration with bKash payment gateway

1.5 Risk Definitions

The following risk severity levels are used throughout this document:

Level	Category	Description
Critical	Business / Legal	Non-compliance with pharmaceutical regulations or data protection laws that could result in shutdown or legal action
High	Security	Breaches or vulnerabilities that expose patient or payment data
Medium	Operational	System downtime or feature failures that degrade user experience significantly
Low	Performance	Minor slowdowns or UI inconsistencies that do not block core functionality

SECTION 2 · OVERALL DESCRIPTION

2 Overall Description

This section provides a high-level overview of the PharmaZen system, including its user base, operating environment, constraints, and assumptions that govern its design and development.

2.1 User Classes and Characteristics

The PharmaZen platform serves four primary user classes, each with distinct access levels and interaction needs:

Guest Visitor	Unauthenticated users who can browse the product catalog, view product details and prices, and create an account. Cannot place orders or upload prescriptions.
Registered Customer	Authenticated users who can manage profiles, upload prescriptions, place and track orders, access order history, and contact pharmacists. Primary revenue-generating user class.
Pharmacist / Staff	Licensed healthcare professionals who verify prescriptions and oversee medication safety flags.
System Administrator	Technical and business administrators who manage all platform operations including user accounts, inventory, promotions, reports, and system configurations.

2.2 User Needs

Based on user class analysis, the following core needs have been identified:

2.2.1 Registered Customers

- Easily search, filter, and find pharmaceutical and health products
- Securely upload and manage prescriptions for prescribed medication orders
- Complete purchases quickly with bKash payment
- Track orders in real time from placement to delivery

2.2.2 Pharmacists / Staff

- Efficiently review and approve or reject uploaded prescriptions
- Flag medications that require counseling or special handling

2.2.3 System Administrators

- Manage product inventory levels, categories, and pricing
- Handle user disputes, returns, and policy enforcement

- View sales analytics reports

2.3 Operating Environment

The PharmaZen website will operate in the following technical environment:

Platform Type	Web application accessible via modern desktop and mobile browsers
Frontend	React.js (Vite) — responsive single-page application targeting screens from 320px to 2560px wide
Backend	Node.js with Express.js REST API, or equivalent server-side framework
Database	PostgreSQL hosted on Neon Tech (serverless)
Cloud Hosting	Vercel (frontend and serverless functions); automatic CI/CD from Git repository
CDN	Vercel Edge Network for static assets, images, and global performance
Browser Support	Chrome 100+, Firefox 100+, Safari 15+, Edge 100+ on desktop and mobile
Payment Gateway	Create mock bKash server for token pass testing, actual Sandbox requires trade license
SSL/TLS	HTTPS enforced across all pages and API endpoints

2.4 Constraints

- **Regulatory Compliance:** The platform must comply with applicable pharmaceutical regulations and digital health guidelines in the operating jurisdiction.
- **Data Privacy:** All user health and personal data must be handled in accordance with applicable data protection regulations (e.g., GDPR, HIPAA or local equivalents).
- **Prescription Verification:** Medications classified as prescription-only must not be dispensed without a valid, verified prescription from a licensed pharmacist.
- **Payment Security:** All payment processing must be handled securely via the bKash payment gateway.
- **Technology Stack:** The development team is constrained to use the agreed-upon technology stack to ensure maintainability and team expertise alignment.
- **Budget & Timeline:** Development must be delivered within the agreed project budget and timeline as defined in the project charter.
- **Third-Party APIs:** Features relying on external APIs (e.g., delivery tracking, payment) are subject to the availability and terms of those third-party providers.

2.5 Assumptions

- All users accessing the platform are assumed to have access to a stable internet connection.
- The platform will initially target a single jurisdiction/country; multi-country expansion is out of scope for v1.0.
- Licensed pharmacists reviewing prescriptions will be pre-vetted and onboarded by PharmaZen management.
- Payment gateway providers will maintain their service agreements and uptime SLAs.
- Product catalog and inventory data will be initially seeded by the PharmaZen operations team.
- Users are responsible for providing accurate health and prescription information; the platform is not liable for fraudulent submissions.
- Third-party delivery partners will be responsible for last-mile logistics, and PharmaZen will integrate only with their tracking APIs.

SECTION 3 · SYSTEM REQUIREMENTS

3 Requirements

3.1 Functional Requirements

Functional requirements define the specific behaviors and capabilities the PharmaZen system must provide.

3.1.1 User Registration & Authentication

ID	Requirement	Description	Priority
FR-01	User Registration	Users can register using an email address and password.	High
FR-02	User Login	Registered users can log in with valid credentials.	High
FR-03	Profile Management	Authenticated users can update personal details (name, address, phone), medical profile, and communication preferences.	High

3.1.2 Product Catalog & Search

ID	Requirement	Description	Priority
FR-07	Product Listings	All products must display name, image, category, price, dosage (if applicable), availability status, and a manufacturer description.	High
FR-08	Search Functionality	Users can search products by name, category, brand, or active ingredient. Search results appear within 2 seconds.	High
FR-09	Advanced Filters	Users can filter results by price range, category, brand, prescription requirement, and in-stock status.	Medium
FR-10	Product Detail Page	Each product has a dedicated page with full description, images, dosage information, side effects, and related products.	High

3.1.3 Prescription Management

ID	Requirement	Description	Priority
FR-13	Prescription Upload	Users can upload prescriptions in PDF, JPEG, or PNG format (max 10 MB) for prescription-required items.	High

ID	Requirement	Description	Priority
FR-14	Prescription Review	Pharmacist staff can view, approve, or reject uploaded prescriptions with mandatory comment fields for rejections.	High
FR-15	Prescription History	Users can access a history of all submitted prescriptions and their current statuses.	Medium

3.1.4 Shopping Cart & Checkout

ID	Requirement	Description	Priority
FR-18	Shopping Cart	Users can add, remove, and update quantities of products in a persistent cart (saved across sessions for authenticated users).	High
FR-19	Checkout Process	Checkout flow includes delivery address selection, shipping method, payment, and order summary review before confirmation.	High
FR-20	Payment Gateway	The system supports payment via bKash mobile wallet. All transactions are processed over HTTPS.	High
FR-21	Order Confirmation	Upon successful payment, the user receives an on-screen confirmation with order details.	High

3.1.5 Order Management

ID	Requirement	Description	Priority
FR-24	Order History	Authenticated users can view all past orders with status, products, total cost, and delivery details.	High
FR-25	Order Tracking	Users can view order status updates on their order history page.	High
FR-26	Order Cancellation	Users can cancel orders if the order has not been paid yet.	Medium

3.1.6 Admin Dashboard

ID	Requirement	Description	Priority
FR-32	Inventory Management	Admins can add, edit, deactivate products, and set low-stock alert thresholds.	High
FR-33	User Management	Admins can view, suspend, or delete user accounts, and manage pharmacist onboarding.	High
FR-34	Order Management	Admins can view all orders, update statuses, initiate refunds, and flag disputes.	High

ID	Requirement	Description	Priority
FR-35	Sales Analytics	Admins can view sales breakdown by medicine, generic name, company, and category.	Medium

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the PharmaZen system must satisfy.

3.2.1 Performance

ID	Requirement	Description	Priority
NFR-01	Page Load Time	All public-facing pages must load within 3 seconds under normal load conditions (up to 500 concurrent users).	High
NFR-02	Search Response	Product search queries must return results within 2 seconds for 95% of requests.	High
NFR-03	Checkout Throughput	The checkout process must support at least 100 concurrent transactions without degradation in response time.	High
NFR-04	Scalability	The system must horizontally scale to support up to 10,000 concurrent users during peak periods (e.g., promotional events).	Medium

3.2.2 Security

ID	Requirement	Description	Priority
NFR-05	Data Encryption	All sensitive data (passwords, health records, payment info) must be encrypted at rest (AES-256) and in transit (TLS 1.2+).	High
NFR-06	Auth Security	Passwords must be hashed using bcrypt (minimum cost factor 12). Plain-text passwords must never be stored.	High
NFR-07	Session Management	User sessions must expire after 30 minutes of inactivity. JWT tokens must have a maximum lifespan of 24 hours.	High
NFR-08	Input Validation	All user inputs must be validated and sanitized server-side to prevent SQL injection, XSS, and CSRF attacks.	High
NFR-09	Vulnerability Scan	The platform must undergo automated vulnerability scans (OWASP Top 10) before each major release.	Medium

3.2.3 Reliability & Availability

ID	Requirement	Description	Priority
NFR-11	Uptime SLA	The platform must maintain a minimum uptime of 99.5% measured on a monthly basis (excluding scheduled maintenance windows).	High

3.2.4 Usability & Accessibility

ID	Requirement	Description	Priority
NFR-16	Responsive Design	All pages must be fully functional and visually consistent across screen sizes from 320px (mobile) to 2560px (desktop).	High
NFR-17	Error Messages	All error messages must be user-friendly, specific, and provide actionable guidance. Technical error codes must not be exposed to end users.	Medium
NFR-19	Onboarding	New users should be able to complete registration and place their first order within 5 minutes without external assistance.	Medium

3.2.5 Maintainability & Compliance

ID	Requirement	Description	Priority
NFR-20	Code Standards	All code must follow the agreed-upon coding standards and pass automated linting checks before merging to the main branch.	Medium

Document Control

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This document is subject to change management procedures. Any modifications must be reviewed, approved, and version-controlled. Changes will be tracked in the project change log.